

Exercise – *Lions versus Hyenas*

“I just can’t wait to be king!”

Young Simba — *The Lion King*

Young Simba finds joy and adventure in the vast savannahs that are his to explore. Amidst his playful endeavors, a favorite game emerges among the lion cubs: “Lions versus Hyenas”, a spirited competition that mirrors the age-old rivalry within their kingdom. To reduce the risk of quarrels among the cubs, Simba’s mother, Sarabi, wishes to split the cubs into two teams, while taking their personal preferences into account.

Each cub i possesses a natural *affinity* to play as either a lion (team A) or a hyena (team B), determined by an integer a_i . A positive a_i means the cub naturally leans towards team A, the lions, while a negative value indicates a preference to be on team B, the hyenas. Additionally, the bonds of friendship among the cubs introduce a *friendship value* $f_{i,j}$ between every pair (i, j) of cubs, reflecting their desire to be on the same team. The friendship value is symmetric ($f_{i,j} = f_{j,i}$) and nonnegative ($f_{i,j} \geq 0$).

In organizing the teams, Sarabi aims to either maximize the *global happiness* of the cubs or minimize their *global dissatisfaction*. When cub i is placed on their preferred team, they add a value of $|a_i|$ to the global happiness. If placed on the opposite team, they subtract $|a_i|$ from the global happiness and add $|a_i|$ to the global dissatisfaction. The friendship value plays a crucial role as well: If two cubs i and j are on the same team, they add $f_{i,j}$ to the global happiness; else they are on different teams, which subtracts $f_{i,j}$ from the global happiness and adds $f_{i,j}$ to the global dissatisfaction. Note that this contribution is considered for each unordered pair $\{i, j\}$, not for both ordered pairs (i, j) and (j, i) . Also note that cubs placed on their preferred team and united friends do not affect the global dissatisfaction.

There is a complication, however, as Simba starts to have romantic feelings for Nala, another lion cub. As Simba is too immature to confront these feelings, Sarabi wants to make sure that Simba and Nala are on different teams. So the challenge for Sarabi is to find an assignment of lion cubs into two teams so as to either achieve a maximum global happiness or a minimum global dissatisfaction, under the constraint that Simba and Nala are on different teams.

Input The first line of the input contains the number $t \leq 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line that contains four integers n b k m , separated by a space. They denote
 - n , the number of lion cubs ($2 \leq n \leq 100$);
 - b , to indicate if Sarabi wishes to maximize the global happiness ($b = 0$) or minimize the global dissatisfaction ($b = 1$);
 - k , the index of Simba ($0 \leq k \leq n - 1$); and

- m , the index of Nala ($0 \leq m \leq n - 1$ and $k \neq m$).
- The following line contains n integers, separated by a space. The i -th integer a_i , for $0 \leq i \leq n - 1$, denotes the natural affinity of the i -th lion cub ($-2^9 \leq a_i \leq 2^9$).
- Each of the following n lines contains n integers, separated by a space. The i -th integer on the j -th line denotes the friendship value $f_{i,j}$ between the i -th and the j -th lion cub ($0 \leq f_{i,j} \leq 2^9$). Recall that $f_{i,j} = f_{j,i}$, for all $0 \leq i, j \leq n - 1$. You may also assume $f_{i,i} = 0$, for all $0 \leq i \leq n - 1$, and $f_{k,m} = f_{m,k} = 0$.

Output For each test case output a single line with a single integer that either denotes the maximum global happiness (if $b = 0$) or the minimum global dissatisfaction (if $b = 1$) achievable, under the constraint that Simba and Nala are on different teams.

Points There are four groups of test sets, worth 80 points in total. For each group of test sets there is also a corresponding hidden test set, worth 5 points. Overall, you can achieve 100 points.

1. For the first group of test sets, worth 25 points, you may assume that $f_{i,j} = 0$ unless $|j - i| = 1$. Furthermore, you may assume $b = 1$, $k = 0$, and $m = n - 1$.
2. For the second group of test sets, worth 25 points, you may assume that $b = 1$ and $a_i = 0$, for all $0 \leq i \leq n - 1$.
3. For the third group of test sets, worth 15 points, you may assume $b = 1$.
4. For the fourth group of test sets, worth 15 points, there are no additional assumptions.

Sample Input

```
4
4 0 0 2
-2 5 -8 1
0 5 0 1
5 0 1 0
0 1 0 4
1 0 4 0
4 1 0 2
-2 5 -8 1
0 5 0 1
5 0 1 0
0 1 0 4
1 0 4 0
4 1 2 3
-2 5 -8 1
0 5 0 1
5 0 1 0
0 1 0 0
1 0 0 0
4 1 0 3
-2 5 -3 1
0 5 0 0
5 0 1 0
0 1 0 4
0 0 4 0
```

Sample Output

```
17
5
3
4
```